

Student Habits vs Academic Performance

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This mini project is a part of the **Data Science and Analytics** subject, guided by **Prof. Tushar Mahore**. The project focuses on applying data analysis techniques to explore real-world patterns and draw meaningful insights from student behavioral data.

1 Introduction

The mini project 'Student Habits vs Academic Performance' analyzes how different daily habits of students affect their academic performance. The goal is to find relationships between these daily habits and exam scores. By studying and visualizing the data, we can see which habits help students get better marks in exams.

2 Objective

To apply the studied analysis methods and use them to understand and show the outcomes of how different student habits affect their academic performance.

3 Dataset Description

The dataset used is **Student_Habits_vs_Academic_Performance.csv**, which contains information about students daily routines and exam results.

It was downloaded from Kaggle: <https://www.kaggle.com/datasets/jayaantanaath/student-habits-vs-academic-performance>.

The dataset contains multiple parameters, but in this project we focus only on those which might have strong impact on student scores.

Key Columns:

- **study_hours_per_day** - Number of hours a student studies daily
- **social_media_hours** - Time spent on social media each day

- `attendance_percentage` - Class attendance percentage
- `sleep_hours` - Average daily sleep hours
- `exam_score` - Final exam score of the student

4 Steps Followed

This section explains the main steps followed during the project - from importing libraries to understanding the results through data visualization.

- **Import Libraries:** We load `ggplot2` and `dplyr` for data handling and visualization.
- **Load Dataset:** Read the CSV file into R using `read.csv()`.
- **Inspect Data:** View sample data using `head()` and checked summary statistics with `summary()` to understand data ranges and distributions.
- **Clean Data:** Remove missing values and duplicate rows using `na.omit()` and `distinct()`.
- **Transform / Prepare Data:** Keep only the important numeric columns for analysis.
- **Descriptive Statistics:** Used `summary()` to calculate mean, median, and range for each numeric variable.
- **Correlation Analysis:** Used `cor()` to find relationships among variables. This helps identify which factors are most related to exam performance.
- **Data Visualization:** Create scatter plots using `ggplot2` to visualize how each habit relates to exam scores:
 - **Study Hours vs Exam Score:** More study time generally leads to better scores.
 - **Social Media Hours vs Exam Score:** Spending more time on social media is linked to lower scores.
 - **Attendance vs Exam Score:** Students with higher attendance tend to perform better.
- **Interpretation & Conclusion:** Observe the key insights and summarize findings using the `cat()` function.

5 Results and Observations

Based on the visualization we did using graphs we observed that:

- Students who study more hours per day generally achieve higher exam scores.
- More time spent on social media negatively affects academic performance.
- High attendance percentage has a positive impact on exam results.
- Adequate sleep helps maintain good concentration and performance.
- Correlation results confirm that study time and attendance are positively correlated with exam score, while social media hours are negatively correlated.

6 Conclusion

This project demonstrates how basic data cleaning, correlation analysis, and visualization can help draw meaningful insights from student data. By analyzing habits like study hours, social media usage, attendance, and sleep, we identified the key factors that influenced exam performance. The study concludes that consistent study, regular attendance, and balanced sleep improve academic success, while excessive social media use can harm it.

Overall, this project highlights the importance of maintaining good habits which result in better scores and shows how simple data analysis in R can help to make conclusions.